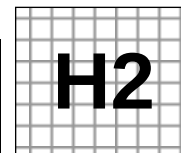


Civics Group	Index Number	Name (use BLOCK LETTERS)
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ST ANDREW'S JUNIOR COLLEGE
2015 JC2 Preliminary Examinations

H2 BIOLOGY

9648/3

Paper 3: Applications and Planning Question

Wednesday

16 September 2015

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your civics group, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination,

1. Attach Question 5 to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Paper 3	
1	/14
2	/13
3	/13
Total	/40
4 (Planning)	/12

This document consists of **14** printed pages.

[Turn over

Answer **all** questions.

1. Patients suffering from Life-threatening Multiple Thrombosis (LMT) are unable to synthesise sufficient amounts of platelet factor IV (PF4), which are released by activated platelets at sites of injury to promote blood coagulation. Platelets are ubiquitous in the blood stream.

Recent advances in genetic engineering have seen preliminary success in cloning human PF4 gene into bacteria for the mass production of PF4 to be used in clinical treatments of LMT.

Fig. 1.1 shows the simplified map of the plasmid, pYH29, used in the cloning of PF4 gene. The selectable markers used are genes that confer resistance to two antibiotics: Kanamycin (KanR) and Clindamycin (ClnR).

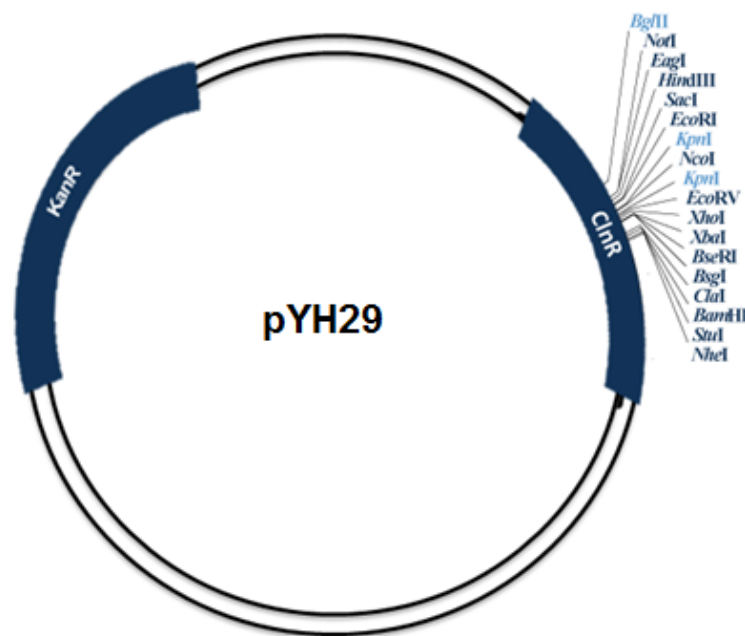


Fig. 1.1

(a) Describe the natural function of a restriction enzyme such as *HindIII*.

.....
[1]

(b) The amount of PF4 gene inserts obtained from cDNA library is normally too little for successful cloning. Suggest one way to increase the amount of the PF4 gene inserts.

.....
[1]

(c) The PF4 gene insert was obtained from cDNA library and cut with *Hind*III (restriction enzyme). Describe and explain the subsequent procedures for cloning the PF4 gene into *E. coli* cells.

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.....[4]

(d) Using both antibiotics, explain how the researchers ensure that only colonies which contain the PF4 gene insert are cultured in large quantities.

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.....[5]

(e) The PF4 gene, when introduced into *E. coli* host cells, produces PF4 that is biologically inactive. Explain why and suggest a possible way to overcome this problem.

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.....[2]

(f) Identify one additional feature of pNY29 plasmid (not shown in Fig. 1.1) and explain why it is needed for the successful production of PF4.

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.....[1]

[Q1 Total: 14]

2(a) X-linked haemophilia is a disease that impairs the body's ability to control blood clotting. The gene affected normally produces Factor VIII, a component of the blood coagulation pathway.

A RFLP marker that is linked to X-linked haemophilia is discovered. Fig. 2.1 shows the two RFLP markers with their restriction sites.

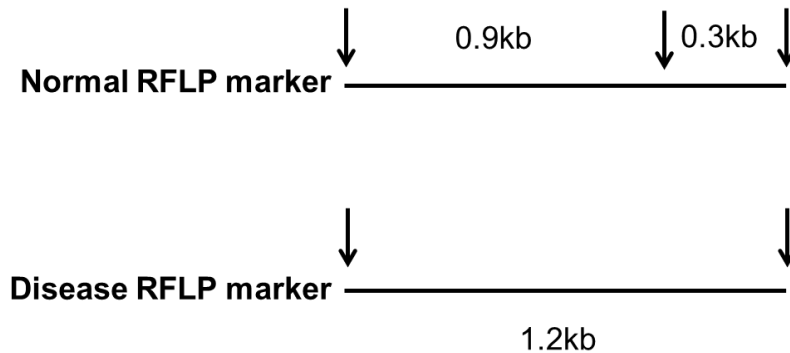


Fig. 2.1

PCR was carried out to amplify the target DNA segment containing the RFLP allele. After which, restriction digestion using *BclI* was done followed by gel electrophoresis to obtain the results shown in the PCR-RFLP analysis in Fig. 2.2. The results are accompanied by the pedigree of a family affected by the disease.

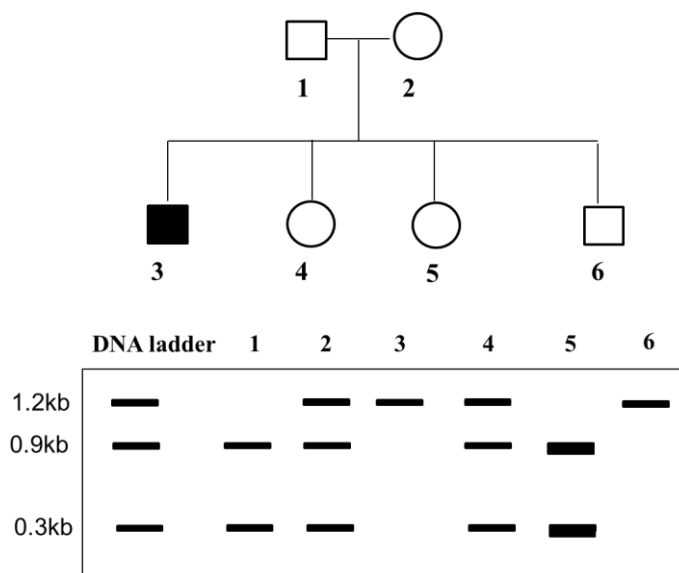


Fig. 2.2

(i) State the role of the DNA ladder seen in Fig. 2.2.

.....
[1]

(ii) On Fig. 2.1, draw the location of the probe on both alleles that would produce an autoradiogram as indicated in Fig. 2.2.

.....[1]

(iii) Contrast and explain the banding patterns of individuals **1** and **5** as seen in Fig. 2.2.

.....
.....
.....
.....[2]

(iv) Other than mutations, account for the observations for individual **6** as seen in Fig. 2.2.

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.....
.....
.....
.....
.....[3]

(b) Suggest why the disease-causing allele causing certain diseases can only be mapped using linkage analysis instead of probing for the disease-causing allele directly.

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.....[1]

(c) DNA profiling is a forensic technique used to identify individuals by characteristics of their DNA.

Explain the basis of application of VNTR in DNA profiling.

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.....[3]

Another method to detect ADA-SCID is through the use of PCR. Fig 2.3 shows the number of DNA molecules made using PCR, starting with one molecule.

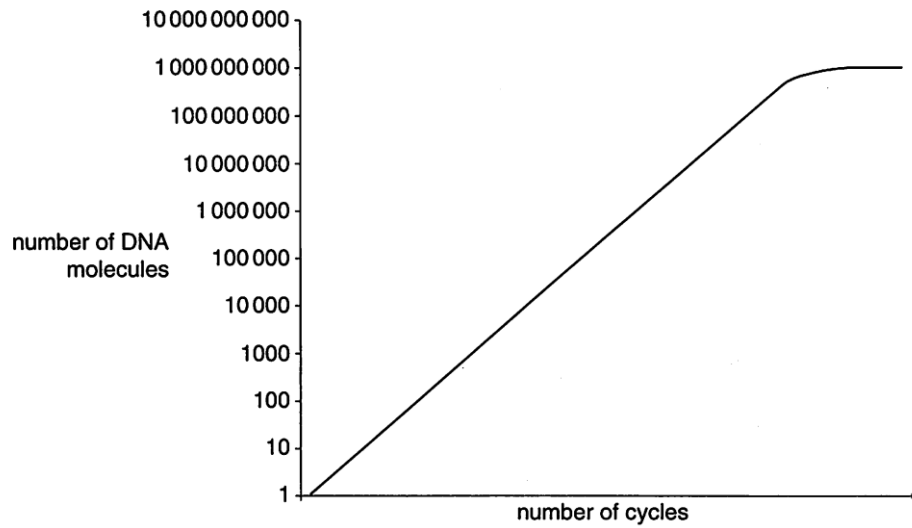


Fig. 2.3

(d) Suggest two reasons for the leveling off of the graph from cycles 17 to 20.

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.....[2]

[Q2 Total: 13]

3(a) Golden rice is a variety of rice (*Oryza sativa*) produced through genetic engineering to produce β -carotene in the edible parts of rice, the endosperm.

Golden rice was created by transforming rice callus tissue with a DNA construct consisting of two β -carotene biosynthesis genes and an antibiotic resistance gene. The genes were placed under the control of a tissue-specific promoter, Glu.

- *psy* (phytoene synthase) from daffodil (*Narcissus pseudonarcissus*)
- *crtI* (carotene desaturase) from the soil bacterium *Erwinia uredovora*
- *Hyg resist* – antibiotic resistance gene from a bacterium

(i) Explain why rice has been genetically modified to produce extra β carotene.

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.....[2]

(ii) Explain what is meant by callus tissue.

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.....[1]

(iii) Suggest the role of the tissue-specific promoter, Glu.

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.....[2]

(iv) Explain the role of the Hyg resistance gene in this procedure.

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.....[1]

(b) The original choice of a *psy* gene from daffodils was made because daffodils produce large amounts of β -carotene in their yellow petals, and because they are monocotyledonous plants, like rice.

In the subsequent years following the invention of Golden Rice, investigations were carried out to see if *psy* genes taken from species other than daffodils would produce greater quantities of β carotene than the first types of Golden Rice.

- *Psy* genes were isolated from the DNA of carrots, daffodils, maize, peppers and tomatoes.
- The genes were inserted into different plasmids.
- The promoter *Glu*, and *crtI* genes from *E. uredovora*, were also inserted into all of the plasmids.
- The five types of genetically modified plasmids were then inserted into different cultures of rice cells.
- The % carotenoid content produced by these rice cells was measured.

Fig. 3.1 show a bar graph of the carotenoid content of calluses transformed with *psy* gene from the five different sources.

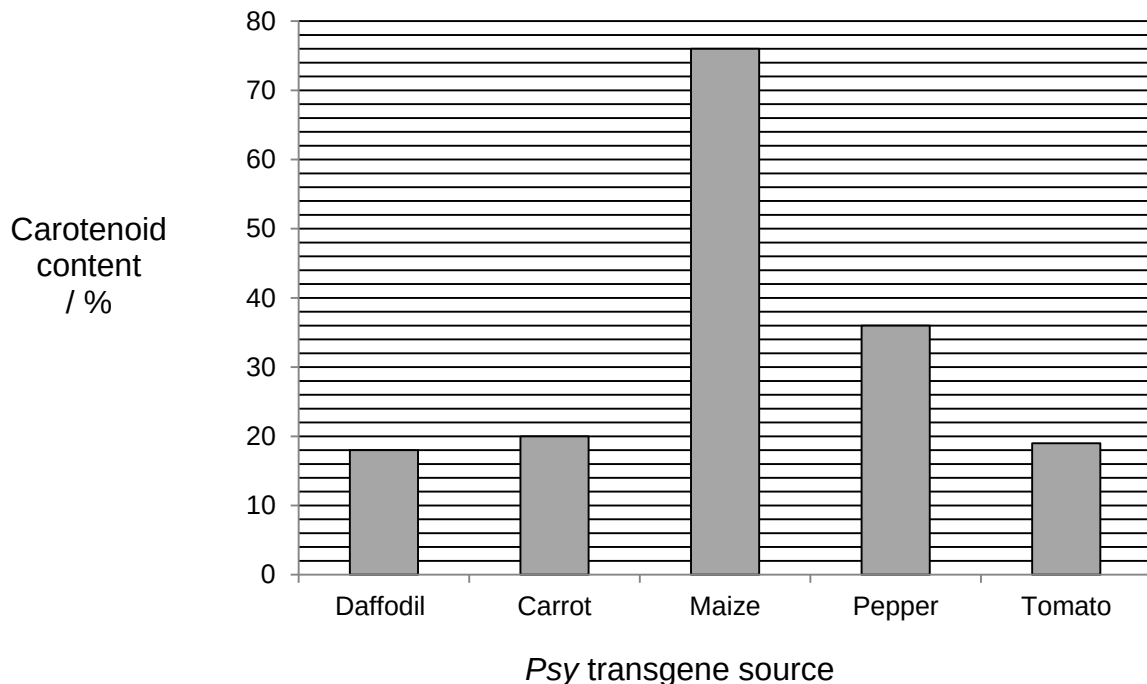


Fig 3.1

(i) Contrast the effectiveness of the *psy* genes of various species in increasing the carotenoid content over that of the daffodil.

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.....[2]

(ii) Suggest reasons for the differences in carotenoid content of the calluses.

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.....[3]

(iii) Describe the possible social implications of growing Golden Rice.

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.....[2]

[Q3 Total: 13]

4. Planning question

You are required to plan, but not carry out, an investigation into the effect of carbon dioxide concentration on the rate of photosynthesis in an aquatic plant, *Elodea canadensis*.

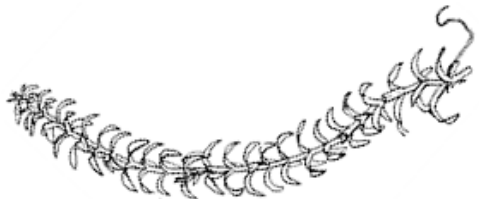


Fig 4.1

Your planning must be based on the assumption that you have been provided with the following equipment and materials:

- pieces of the *Elodea* plant, each with varying lengths exceeding 100 mm long shown in Fig. 4.1
- plasticine (to anchor *Elodea* plant in water)
- an unlimited supply of filtered pond water through which air has been bubbled for at least 24 hours
- an unlimited supply of 0.1 mol dm^{-3} sodium hydrogen carbonate solution
- boiling tubes
- bungs to fit the test-tubes with one L-shaped glass tube passing through
- pieces of capillary tube each 300 mm long, with a bore diameter of 1 mm
- beaker of coloured liquid
- rubber connecting tubings to connect L-shaped glass tubes or capillary tubes
- retort stand with clamps
- scalpel
- ruler marked in mm
- stop watch
- thermometer
- large glass beaker
- supply of hot and cold water
- a lamp at one end of an otherwise unlit bench 2 metres long
- a room which can be made dark
- thick sheet of clear glass (heat shield)

Your plan should:

- show an explanation of the theory to support your practical procedure
- describe the method with scientific reasoning so that the results are as accurate and reliable as possible
- be illustrated by relevant diagrams, if necessary
- show how you will record your results and the proposed layout of results table and graphs,
- include reference to safety measures to minimise any risks associated with the proposed experiment,
- use the correct technical and scientific terms.

[Q4 Total: 12]

[illegible]

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 **(a)** There are two common forms of severe combined immunodeficiency (SCID). Contrast between adenosine deaminase (ADA) deficiency and X-linked SCID. [5]
- (b)** Using one named example of an adult stem cell, describe its normal role and how it could be used in gene therapy against SCID using a retrovirus. [8]
- (c)** Discuss the goals and benefits of the human genome project. [7]

[Total: 20]